GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Signals & Systems — Official GATE ECE Syllabus

Paper Code: EC • Section 2: Networks, Signals and Systems (7 - 9 Marks)

Weightage: 7 - 9 Marks • **Subject Code:** SS • **Key Focus:** LTI Systems, Transforms, Sampling

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: CONTINUOUS-TIME AND DISCRETE-TIME SIGNALS

- Signal classifications: Continuous-time (CT) vs. Discrete-time (DT), deterministic vs. random, causal vs. non-causal.
- Periodic vs. aperiodic signals: Fundamental period calculation for sum of continuous and discrete sinusoids.
- Energy signals, Power signals, and neither energy nor power signals. Energy Spectral Density (ESD) and Power Spectral Density (PSD).
- Even and odd components of real and complex signals. Orthogonality of signals.
- Elementary signals: Unit step function, unit impulse function (Dirac delta and Kronecker delta), ramp function, signum function, rect and sinc functions.
- Transformations of independent variable: Time shifting, time scaling, time reversal.

SECTION: LINEAR TIME-INVARIANT (LTI) SYSTEMS

- System properties: Linearity (homogeneity and additivity), Time-invariance, Causality, Stability (BIBO stability criteria).
- Memory (static vs. dynamic), Invertibility and inverse systems.
- Impulse response h(t) and h[n]. Convolution integral and convolution sum: Properties (commutative, associative, distributive).
- Graphical convolution, duration of convolved signals (L_total = L1 + L2 - 1).
- Step response as integral/sum of impulse response. Interconnection of LTI systems (cascade and parallel).
- Eigenfunctions and eigenvalues of LTI systems: Complex exponentials as eigenfunctions.

SECTION: FOURIER SERIES AND CONTINUOUS-TIME FOURIER TRANSFORM (CTFT)

- Continuous-Time Fourier Series (CTFS): Trigonometric and Exponential Fourier series, Dirichlet's conditions.
- Fourier coefficients, line spectrum, Parseval's power theorem for periodic signals.
- Continuous-Time Fourier Transform (CTFT): Direct and inverse transform equations.
- Properties of CTFT: Linearity, time shifting, frequency shifting, time scaling, duality property, time differentiation and integration, frequency differentiation.
- Convolution property and modulation (multiplication) property. Parseval's energy relation.
- Transform pairs of standard functions: Impulse, unit step, complex exponential, rect, sinc, Gaussian pulse.

SECTION: SAMPLING THEOREM AND SIGNAL RECONSTRUCTION

- Sampling theorem for low-pass and band-pass signals. Nyquist rate and Nyquist interval.
- Ideal impulse train sampling, flat-top sampling, and natural sampling.

- Aliasing phenomenon and anti-aliasing filters. Guard bands.
- Signal reconstruction from samples using ideal low-pass interpolation filter (sinc interpolation).

SECTION: LAPLACE TRANSFORM AND Z-TRANSFORM

- Laplace Transform: Bilateral and unilateral transforms, Region of Convergence (ROC) properties.
- Poles and zeros, stability and causality conditions in terms of ROC (ROC includes jw axis for stability; ROC right of rightmost pole for causality).
- Properties of Laplace transform, initial and final value theorems, inverse Laplace transform using partial fractions.
- Z-Transform: Definition, bilateral and unilateral Z-transforms, ROC and its geometric properties in the z-plane.
- Causality and stability criteria in Z-domain (unit circle must lie in ROC for stability; ROC outside outermost pole for causality).
- Inverse Z-transform methods: Power series expansion, partial fraction expansion, contour integration.

SECTION: DISCRETE-TIME FOURIER TRANSFORM & FFT BASICS

- Discrete-Time Fourier Transform (DTFT): Periodicity of DTFT (2*pi periodic), properties, symmetry relations.
- Discrete Fourier Transform (DFT): N-point DFT and IDFT definitions, circular shift, circular convolution vs. linear convolution.
- Fast Fourier Transform (FFT): Decimation-in-Time (DIT) and Decimation-in-Frequency (DIF) radix-2 FFT algorithms, computational complexity reduction (O(N log2 N) vs O(N^2)).
- Introduction to digital filters: FIR and IIR filter comparison, linear phase conditions.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- Signals and Systems — Alan V. Oppenheim, Alan S. Willsky & S. Hamid Nawab (Pearson)
- Signals and Systems — Simon Haykin & Barry Van Veen (Wiley)
- Digital Signal Processing — John G. Proakis & Dimitris G. Manolakis

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- Nyquist rate for product of signals: m1(t)*m2(t) has Nyquist rate = 2*(fm1 + fm2).
- In Z-transform, if a system is causal and stable, all poles must lie strictly INSIDE the unit circle (|z| < 1).
- Initial and final value theorems apply ONLY when poles strictly satisfy left-half s-plane / interior z-plane stability.